# ARM7 Load–Store Architecture

## 1 Load–Store Architecture Overview

ARM7 follows a **load–store architecture**, where only load and store instructions access memory. All arithmetic and logical operations are performed using registers.

- Memory accessed only by **LDR** and **STR**
- ALU operates only on register data
- Reduces memory access latency
- Improves execution speed and pipelining

## 2 Instruction Control and Addressing Units

### Instruction Decoder and Logic Control

- Fetches and decodes instructions
- Determines operation type and operand registers
- Generates control signals for ALU, registers, and memory
- Coordinates internal and external processor activities

### Address Register

- Holds the 32-bit memory address
- Sends address to the address bus
- Used during instruction fetch and data access

### Address Incrementer

- Increments address value by 4
- Supports sequential instruction execution
- Works with 32-bit (4-byte) ARM instructions

## 3 Register Bank

The register bank is a key performance feature of ARM7, allowing most operations to be completed without accessing memory.

- Contains thirty-one 32-bit general-purpose registers
- Includes six status registers
- Stores operands, intermediate values, and final results
- Minimizes slow memory operations

## 4 Execution Units

### Booth's Multiplier

- Performs fast multiplication
- Efficient for signed and large operands
- Reduces number of arithmetic steps

### Barrel Shifter

- Performs shift and rotate operations
- Can shift by multiple bits in a single cycle
- Supports logical, arithmetic, and rotate operations

### Arithmetic Logic Unit (ALU)

- 32-bit ALU
- Performs arithmetic and logical operations
- Updates condition flags (N, Z, C, V)

## 5 Data Transfer Registers

### Write Data Register

- Holds data to be written to memory or I/O
- Used during store operations

### Read Data Register

- Holds data read from memory or I/O
- Supplies data to registers after load operations

## 6 Bus Structure and Parallelism

ARM7 achieves high performance through extensive use of parallel data paths and dedicated buses.

### Bus Organization

- Non-multiplexed address and data buses
- Separate read and write data buses
- Dedicated operand buses A and B for ALU inputs
- Separate ALU output bus

### Advantages of Multiple Buses

- Enables parallel data transfer
- Eliminates need for temporary registers
- Allows overlapping of read, execute, and write operations
- Enhances execution speed and throughput

ARM7 can write the result of one instruction while simultaneously fetching operands for the next instruction.

## 7 Key Takeaways

ARM7 combines load–store architecture, powerful execution units, and parallel bus design to achieve high performance with low power consumption.

- Load–store design minimizes memory access
- Dedicated execution units improve speed
- Multiple buses enable true parallelism
- Architecture is highly pipeline-friendly